Microsoft 365

MD-100 Windows Client

Module 11 : Troubleshoot hardware and drivers

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Module agenda

- Troubleshoot device driver failures
- Explore physical hardware troubleshooting
- Troubleshoot physical failures on Windows clients

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Lesson 1: Troubleshoot device driver failures

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Lesson 1: Troubleshoot device driver failures

- Examine tools for managing and troubleshooting device drivers
- Use driver roll back
- Manage signed drivers
- Stage device drivers
- Restrict device installation

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Examine tools for managing and troubleshooting device drivers

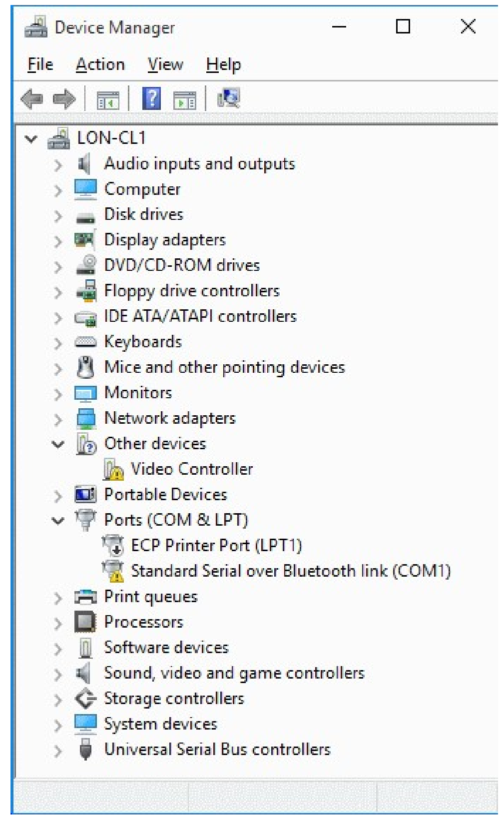
Use Device Manager to:

- View devices, their drivers, details, settings, and events
- Enable and disable devices
- Install, update, uninstall, and roll back device drivers
- Troubleshoot device issues
- Manage devices locally only

Other tools include:

- Windows PowerShell
- DevCon.exe tool

Remote Desktop or Windows PowerShell remoting



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Use driver roll back

Nondestructive operation, but requires restart

Reinstalls previous version of device driver:

- Not available if device driver has never been updated
- Only active and functional drivers are backed up

Supports one level of rollback:

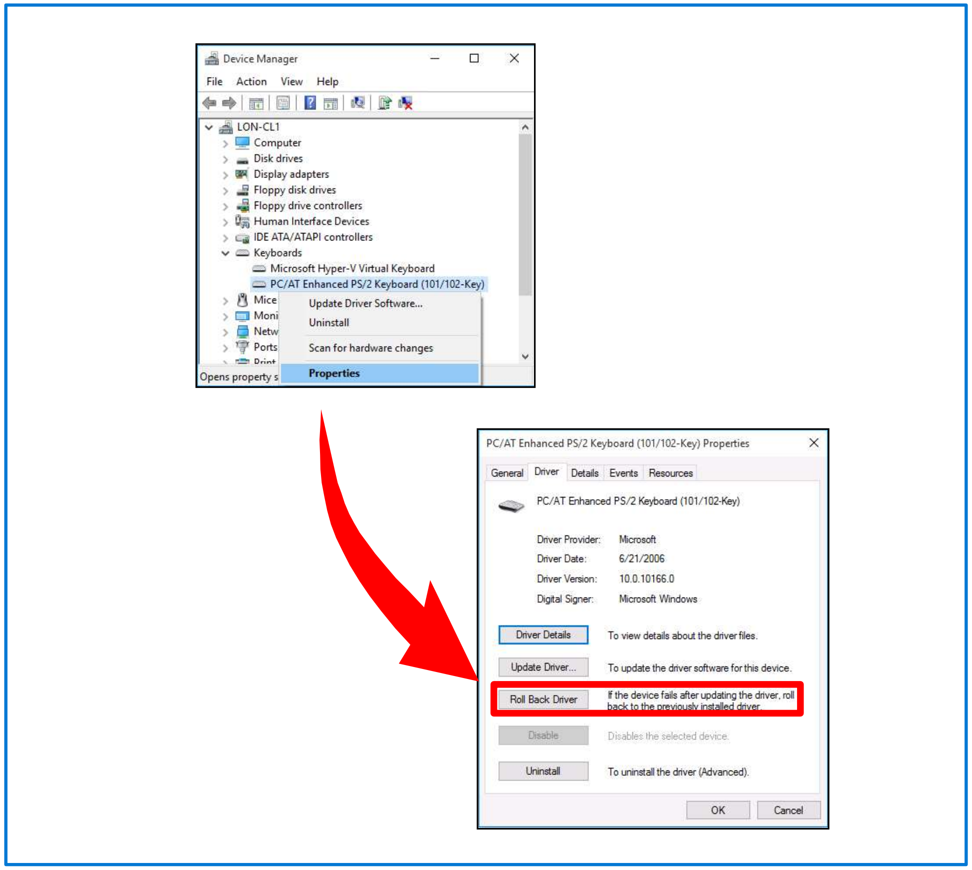
- If you perform a rollback, you cannot roll back again

Driver Roll Back is not available for printers

Multifunction devices are handled on an individual function basis (printer, scanner)

Can be performed from safe mode:

- If malfunctioning driver is preventing normal operation



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Manage signed drivers

Windows 10:

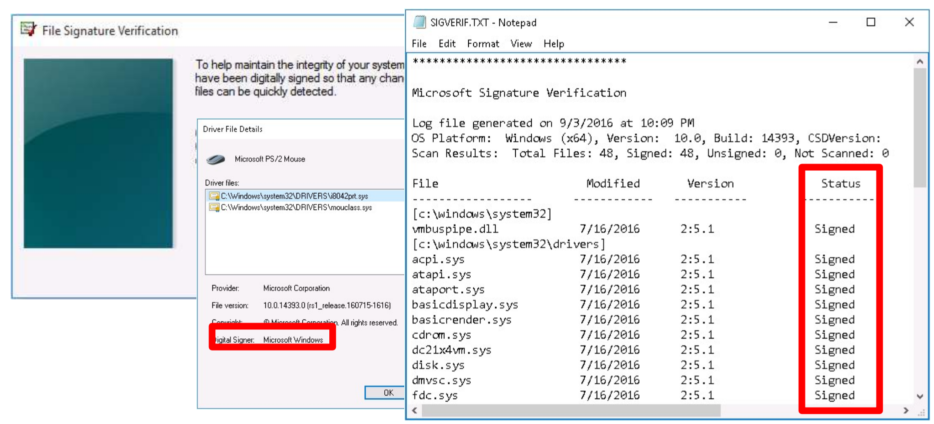
- 32-bit warns you if a driver is not signed
- 64-bit requires signed drivers

Windows 11:

- 64-bit only, requires signed drivers

Driver signing does not modify driver functionality

- Run Sigverif.exe to scan for unsigned drivers.



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Stage device drivers

Benefits of Staging Driver Packages

- Allows only known authorized drivers to run
- Improves security
- Reduces support costs
- Better user experience

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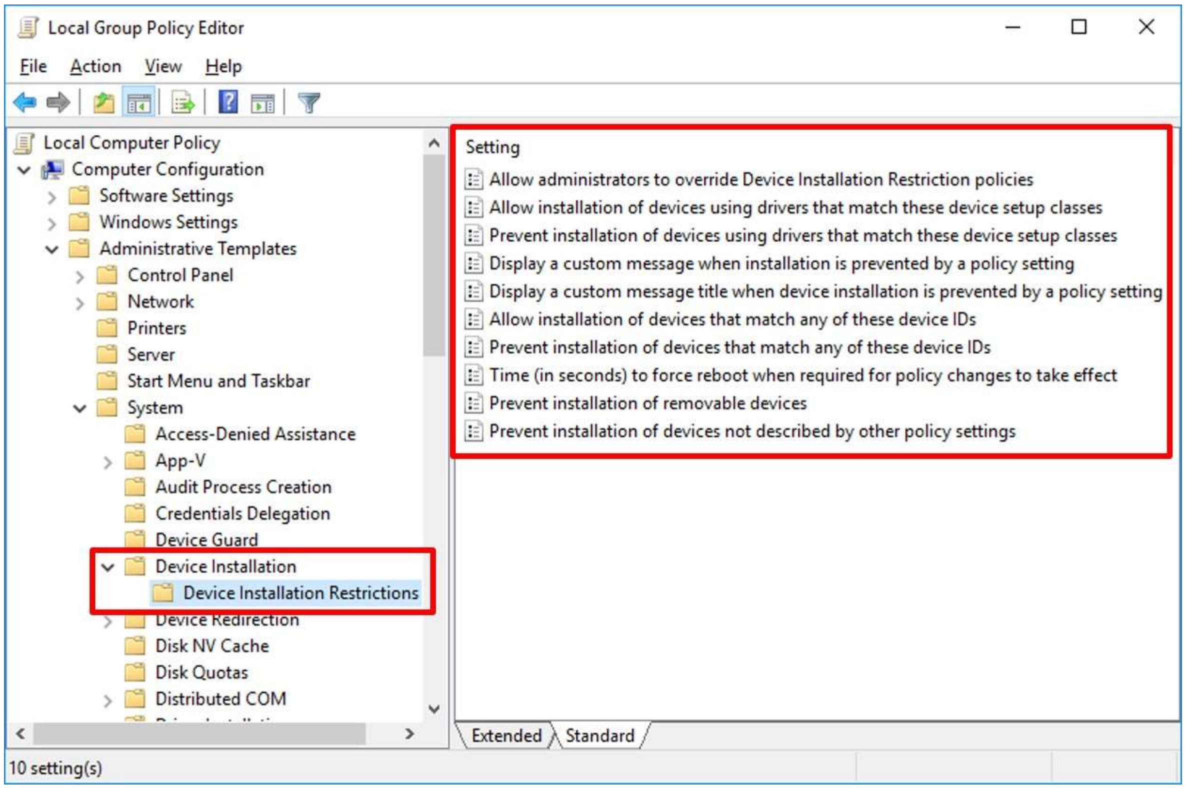
Restrict device installation



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Knowledge Check

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Lesson 2: Explore physical hardware troubleshooting

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Lesson 2: Explore physical hardware troubleshooting

- Identify hardware-related problems
- Troubleshoot USB devices
- Troubleshoot wireless devices
- Gather hardware information
- Use best practices to troubleshoot hardware issues

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Identify hardware-related problems

Physical Issue

- Moving parts (fans, power supplies, hard disk drives) prone to fail
- Excessive dust impairing heat dissipation

Driver Issue

- Version incompatibility
- Defective driver
- Incorrect driver assigned

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Troubleshoot USB devices

Controlling USB Devices

- Prevent users from installing any device.
- Allow users to install only devices on an approved list.
- Prevent users from installing devices that are on a prohibited list.
- Deny read or write access to removable devices or removable media.

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Troubleshoot wireless devices

Troubleshooting Wireless Devices

- WiFi/Bluetooth Enabled
- Airplane mode
- Discovery
- Pairing
- Device proximity
- External interference

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Gather hardware information

- Event Viewer
- System Information Tool
- Device Manager
- Windows Powershell
- Windows Memory Diagnostics tool
- Central Device Management Inventory

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Use best practices to troubleshoot hardware issues

- Physical Connection?
- Compatible Device?
- Review Event Viewer logs
- Recent driver change?
- Update device firmware/BIOS

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Knowledge Check

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Lesson 3: Troubleshoot physical failures on Windows clients

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Lesson 3: Troubleshoot physical failures on Windows clients

- Examine factors when replacing devices
- Explore common vulnerabilities on hardware devices
- Examine guidelines for replacing hardware
- Diagnose Memory Problems
- Diagnose and troubleshoot disk problems

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Examine factors when replacing devices

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Explore common vulnerabilities on hardware devices

- Drives
- PSUs
- Fans
- CPU/GPU
- Memory
- Battery

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Examine guidelines for replacing hardware

- Identify root cause
- Warranty concerns
- Static electricity
- User interruption

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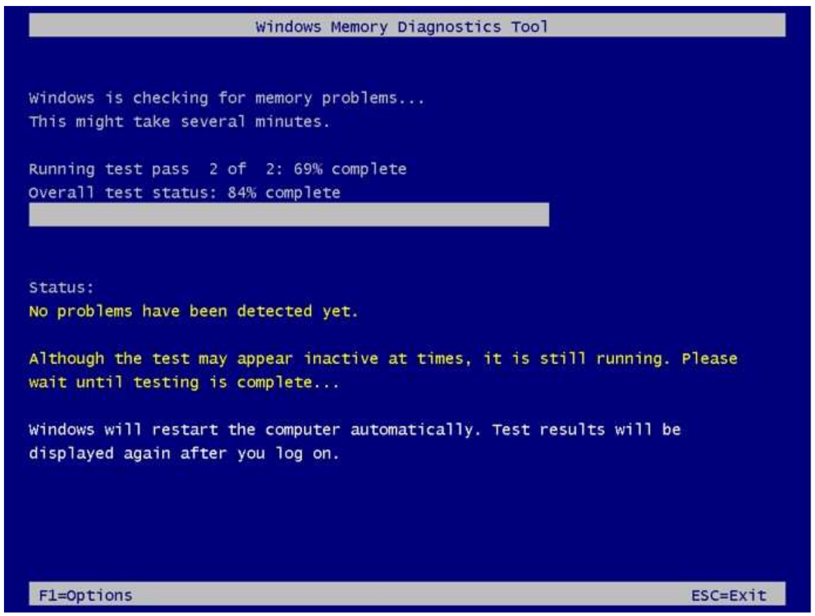
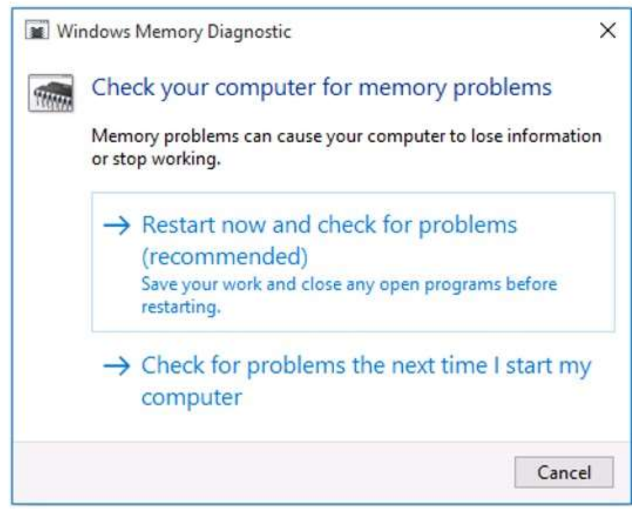
Diagnose Memory Problems

Windows Memory Diagnostics Tool





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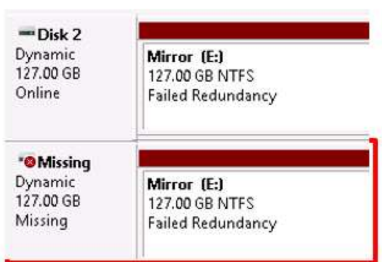
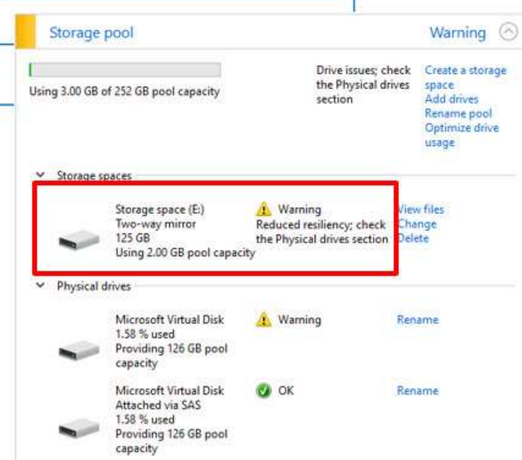
Diagnose and troubleshoot disk problems

You can use following redundant storage types:

Type	Disks required	Number of disks that can fail without loss of data
Mirrored volume	2	1
Parity	3 or more	1
Two-way mirror	2	1
Three-way mirror	5 or more	2





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Knowledge Check

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Practice Labs

Recovering Windows by using a Restore Point

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Module Review

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